iii. Similarly this type of fault is most frequently occurs in the power system.
iv. Their chances of appearance in the power system 70%.

2. Phase-to-Phase fault

i. It is also called line-to-line fault.
ii. It occurs when two conductors are short circuited.
iii. Such type of fault rarely occurred on the power system.
iv. Their chance of appearance is hardly 15% in the power system.

3. Two phases to ground fault

i. It is also called Line-to-line-to-ground fault (L-L-G).
ii. In this type of fault breakdowns of insulation between two phases and earth occur.
iii. It is the most severe type of fault but rarely occurs in the power system.
iv. Their chance of occurrence is hardly 10%.

4. Phase to phase and third Phase to ground fault

i. It is the combination of phase to phase and phase to phase to ground fault.
ii. Such types of fault occur due to the breakdown of insulation between two phases and simultaneous breakdown of insulation between the third phase and earth.
iii. The chance of such type of fault is hardly 2% to 3%.

10.4 General defects in machines

According to the supply, machines are classified into two major types.

1. DC machines
2. AC machines

General defects and its remedy for DC machine are given below.

In DC machine, end connection and poles are very important. In DC machines terminal should be connected correctly (positive terminal to positive terminal and negative terminal to negative terminal), otherwise DC machines does not work.

10.4.1 General defects and its remedies for DC machines

Defects	Reasons	Remedies
1. When an electric supply is given to the DC motors, it is not running	1. Low voltage.	Check each and every phase supply voltage and measure the required voltage value and correct it.
	2. Over load.	Reduce the over load
	3. Fault occurs in winding coils.	Check the winding coils and correct it

BEE_ENG_Chapter_10.indd 157 12/27/2019 10:39:26 AM